Decision Record: Reviewer Comments and the Per-Tonne Note

What was merged, what was not, and why

1 Merged into the report

Everything below is now in `aluminium_report_downstream.tex`. Each item is either arithmetic derived from data already in the report, or a claim with a citation that has been independently verified.

Comment	Where	What was added
Is causation established?	§4.1	A paragraph stating that the evidence is association, not an identified causal effect: no counterfactual, three years of ERP against twelve of ASI, construction cycle uncontrolled. Two overclaiming sentences in §4.6 and §4.7 were rewritten.
Effect of a rise in metal cost	§4.7, App. B.4	$\Delta V/V = -[m\theta/(1-m)]\hat{p}$. With $m = 0.831$ the amplification factor is **4.92**; at $\theta = 60$% a 1% rise in metal price removes ≈3% of downstream value added and the 7.5% duty over a fifth of it. Upstream the figure is nil.
Employment per tonne	§4.3, App. B.4	**41 persons per thousand tonnes upstream against 210–262 downstream** (5.1–6.4), with the exact decomposition $3.14 \times 1.81 = 5.7$ linking it to the per-rupee ratio.
Export mix (from the note)	§3.1	India's exports are **61.4% primary metal**; China's are **2.8%**. India imported US$4.1bn of finished aluminium goods in FY2025–26, a quarter at low or zero duty. Sourced to GTRI, added to the bibliography.
Actual tax or a proxy?	§4.7	An explicit statement that corporate tax is an *estimate* of the tax base, not receipts; axis and table labels renamed accordingly.
"Control" is wrong for the UK	§3.3	Rewritten as a non-preferential (MFN) benchmark holding product basket and estimator constant, with the explicit note that no further causal weight is placed on it.
Price gap not shown to be tariff-driven	§3.1	Softened: the duty is "one mechanism through which such a spread can be sustained", not its demonstrated cause, with the missing decomposition named.

2 Deliberately not merged

1. **The 35:1 employment ratio.** It compares the direct workforce of the primary smelters alone (4.6 times smaller than the ASI's count for NIC 2420) with the direct, indirect and unorganised workforce downstream (1.7 times larger than the ASI's count for the fabrication classes). Those two mismatches multiplied by the like-for-like ratio of 5.7 give $\approx$45, which is where the figure comes from. The report uses 5.1–6.4 and explains the larger number in a footnote, so that a reader who has seen it elsewhere is not left with an apparent contradiction.

2. **The stylised 2,000-tonne table.** Its parameters contradict the real figures cited two paragraphs later in the same note (90 workers per thousand tonnes upstream, against 9–10 in the same document). Two of its four summary percentages are wrong: 150 against 180 workers is 83%, not 67%; $70 against $25 per tonne is 180% more, not 40%. And its value-added-per-worker figures imply downstream is *less* labour-intensive per rupee—the opposite of what the ASI shows.

3. **"Immediate" contract-worker adjustment.** Twelve annual observations against three years of ERP estimates identify no lag structure, and the contract-share series in fact moves gradually. The claim has been removed rather than defended; the composition finding that replaces it needs no timing assumption.

4. **The industry's specific asks.** Drafted text exists but the asks and their ordering are yours to set, not mine to infer from the report. Say the word and it goes in.

5. **Stakeholder impact analysis.** A reasoned draft exists. It is argumentation rather than evidence, and belongs in the Conclusion, which is outside the section rebuilt so far — flagging it rather than inserting it unilaterally.

3 Outstanding — needs data

1. **Bilateral imports, HS 7604–7616 at eight digits, by partner, five years** (DGCI&S or UN Comtrade). Would answer "which Chinese products displace Indian ones", which the tariff files cannot — they carry rates, not volumes. It would also put the trade-remedy argument in §3.2 on evidence rather than assertion. *Highest value of the three.*

2. **Cost sheets from two or three representative fabricators**, giving metal as a share of total input cost. This is θ in the amplification result, and the only unmeasured parameter in the strongest quantitative claim in Section 4. It would replace a range with a number.

3. **LME cash settlement, Indian ex-works prices, freight, insurance and physical premia** over a stated window. Would let the price-gap claim be made properly instead of softened.